Practical No.1

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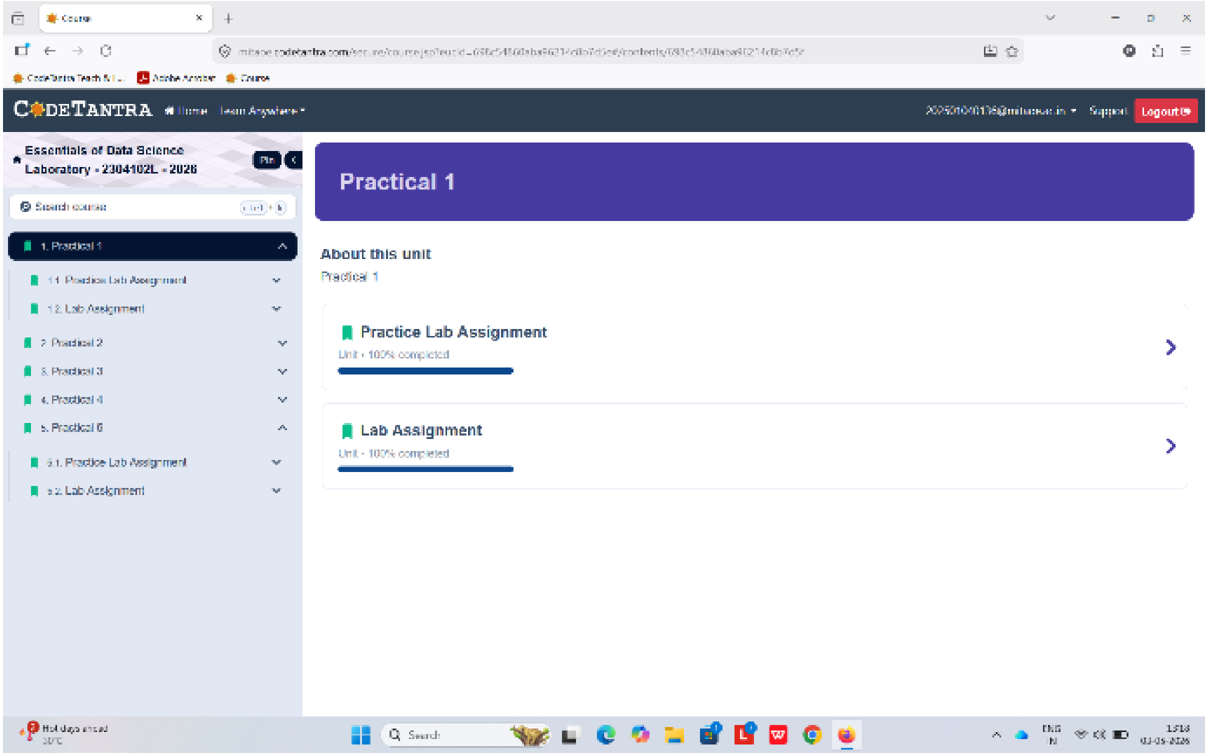

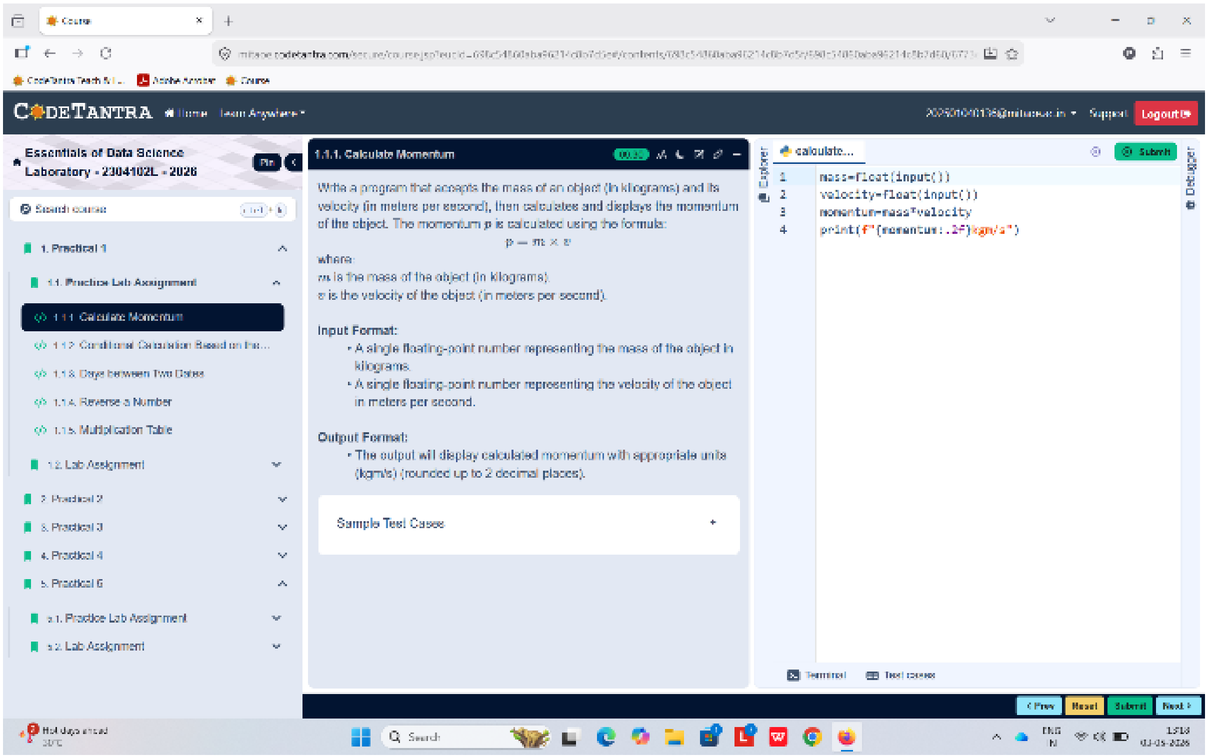

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1.1.2. Conditional Calculation Based on the Number o...

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:

$1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n.

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

```python
import math
n=int(input())
if n>=1 and n<=9:
    print(n*n)
elif n>=10 and n<=99:
    print(f"{math.sqrt(n):.2f}")
elif n>=100 and n<=999:
    print(f"{math.cbrt(n):.2f}")
else:
    print("Invalid")
```


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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

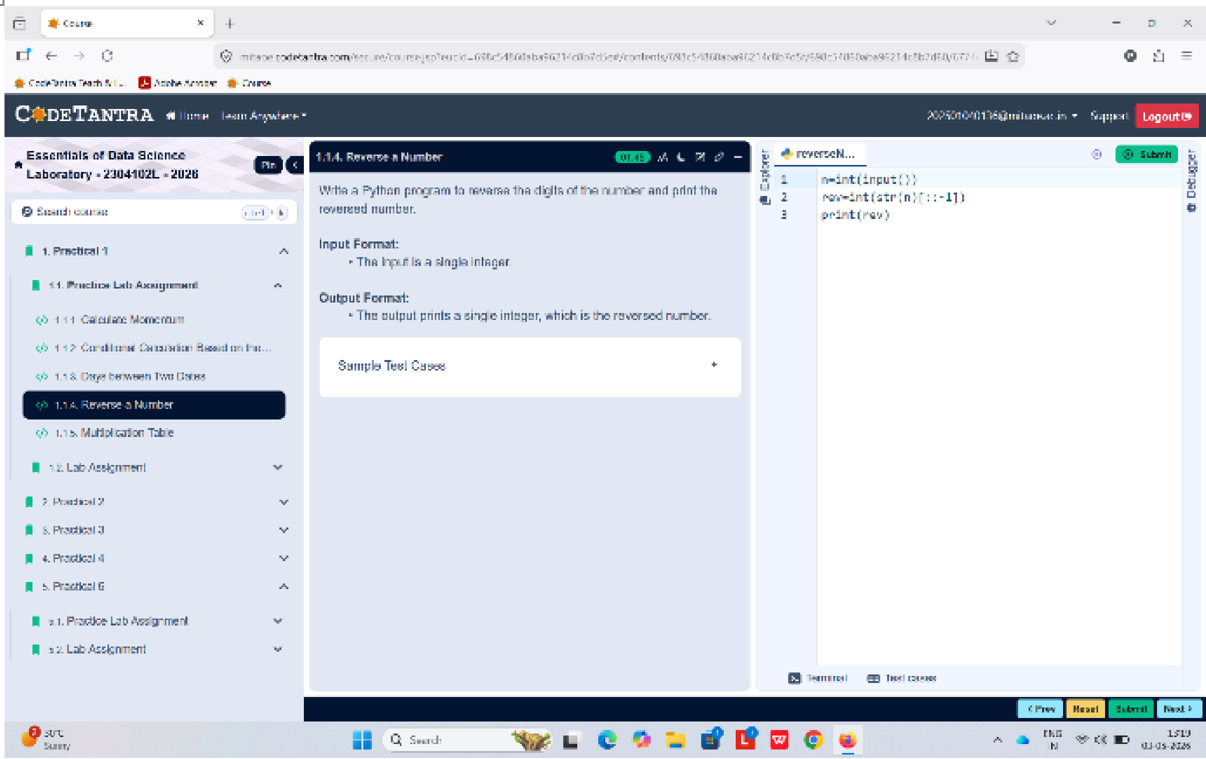
- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

```python
from datetime import date
d1=input()
d2=input()
y1,m1,day1=map(int,d1.split('-'))
y2,m2,day2=map(int,d2.split('-'))
date1=date(y1,m1,day1)
date2=date(y2,m2,day2)
print((date2-date1).days)
```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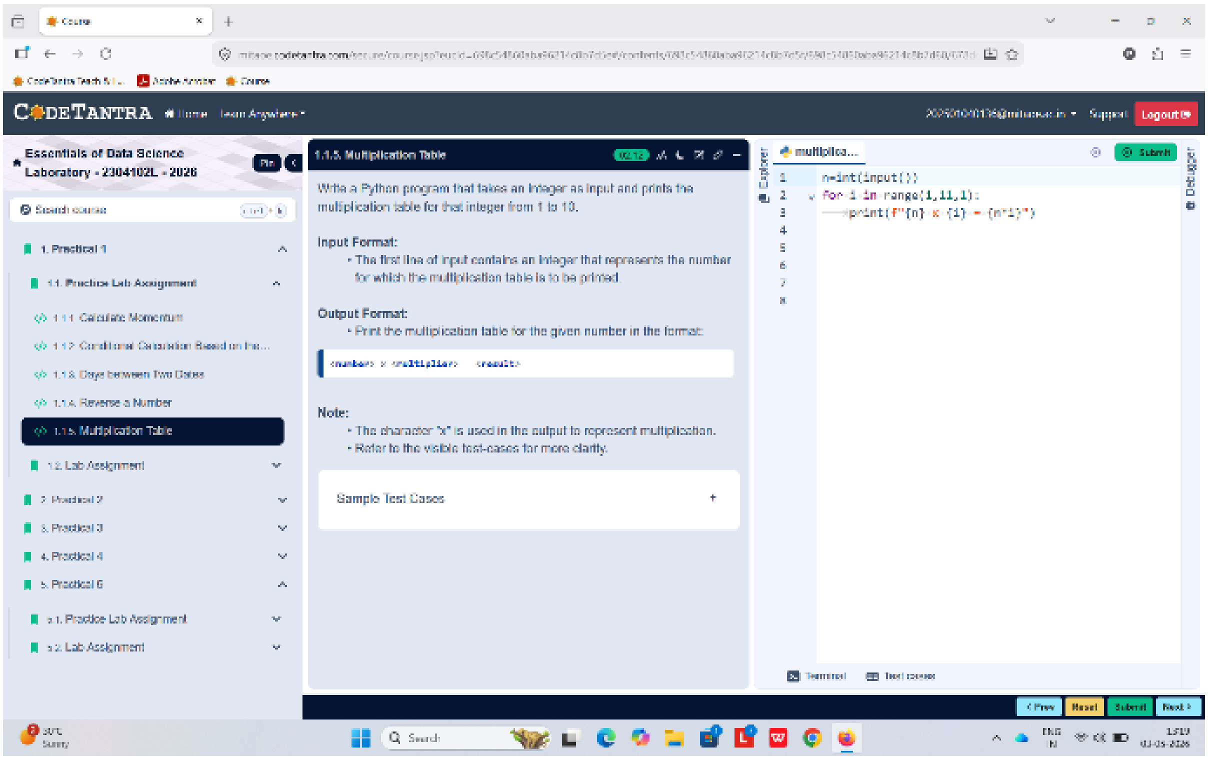
1.1.4. Reverse a Number
Write a Python program to reverse the digits of the number and print the reversed number.
Input Format:
• The input is a single integer.
Output Format:
• The output prints a single integer, which is the reversed number.
Sample Test Cases
n=int(input())
rev=int(str(n)[::-1])
print(rev)
Submit
Logout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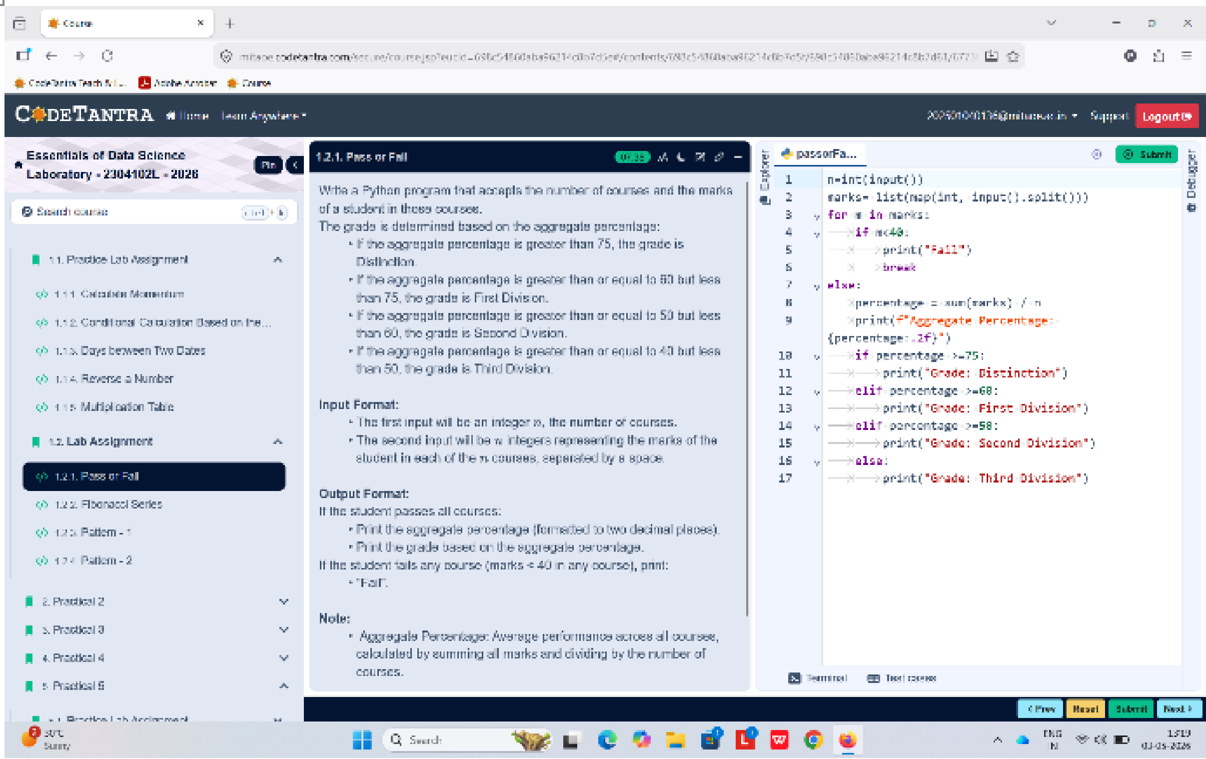
1.1.5. Multiplication Table
Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.
Input Format:
• The first line of input contains an integer that represents the number for which the multiplication table is to be printed.
Output Format:
• Print the multiplication table for the given number in the format:
<number> x <multiplier> <result>
Note:
• The character "x" is used in the output to represent multiplication.
• Refer to the visible test-cases for more clarity.
Sample Test Cases
n=int(input())
for i in range(1,11,1):
print(f"{n} x {i} = {n*i}")
Submit
Logout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1.2.1. Pass or Fail
Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:
• If the aggregate percentage is greater than 75, the grade is Distinction.
• If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
• If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
• If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.
Input Format:
• The first input will be an integer n, the number of courses.
• The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.
Output Format:
If the student passes all courses:
• Print the aggregate percentage (formatted to two decimal places).
• Print the grade based on the aggregate percentage.
If the student fails any course (marks < 40 in any course), print:
• "Fail".
Note:
• Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.
passorFa...
Submit
n=int(input())
marks= list(map(int, input().split()))
for m in marks:
if m<40:
print("Fail")
break
else:
percentage = sum(marks) / n
print(f"Aggregate Percentage: {percentage:.2f}")
if percentage >=75:
print("Grade: Distinction")
elif percentage >=60:
print("Grade: First Division")
elif percentage >=50:
print("Grade: Second Division")
else:
print("Grade: Third Division")
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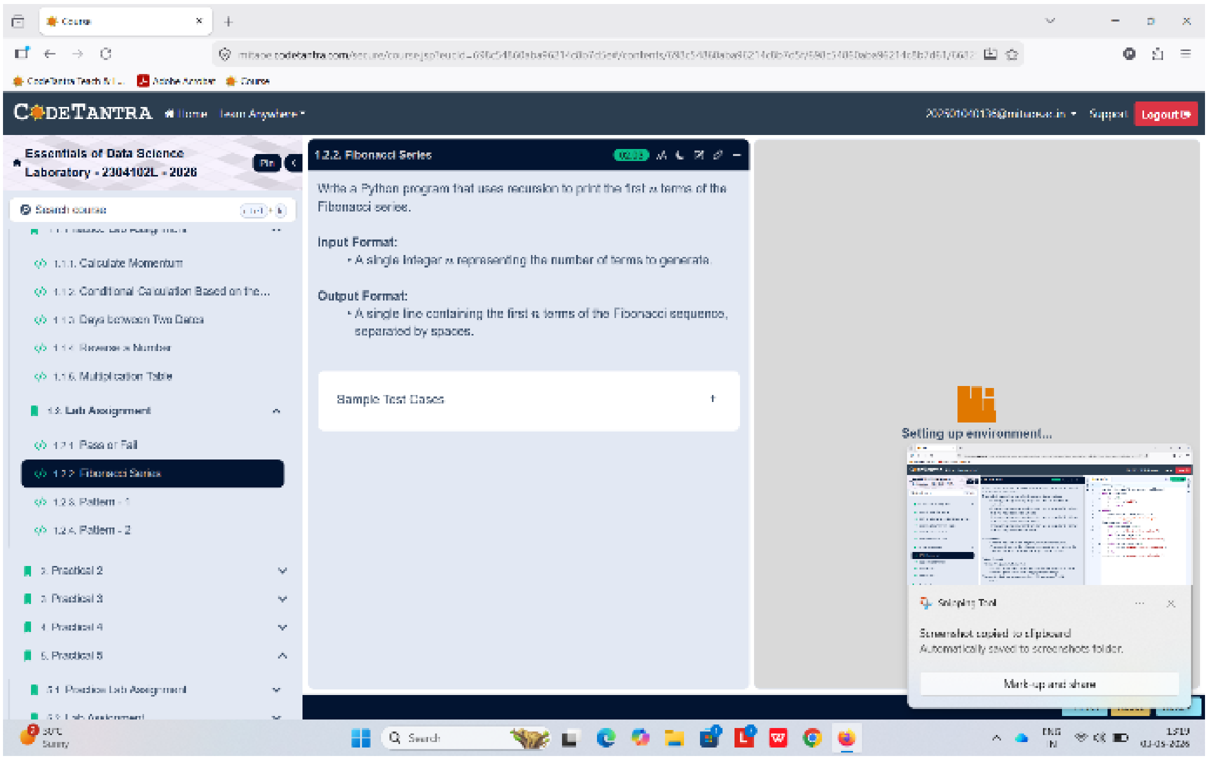
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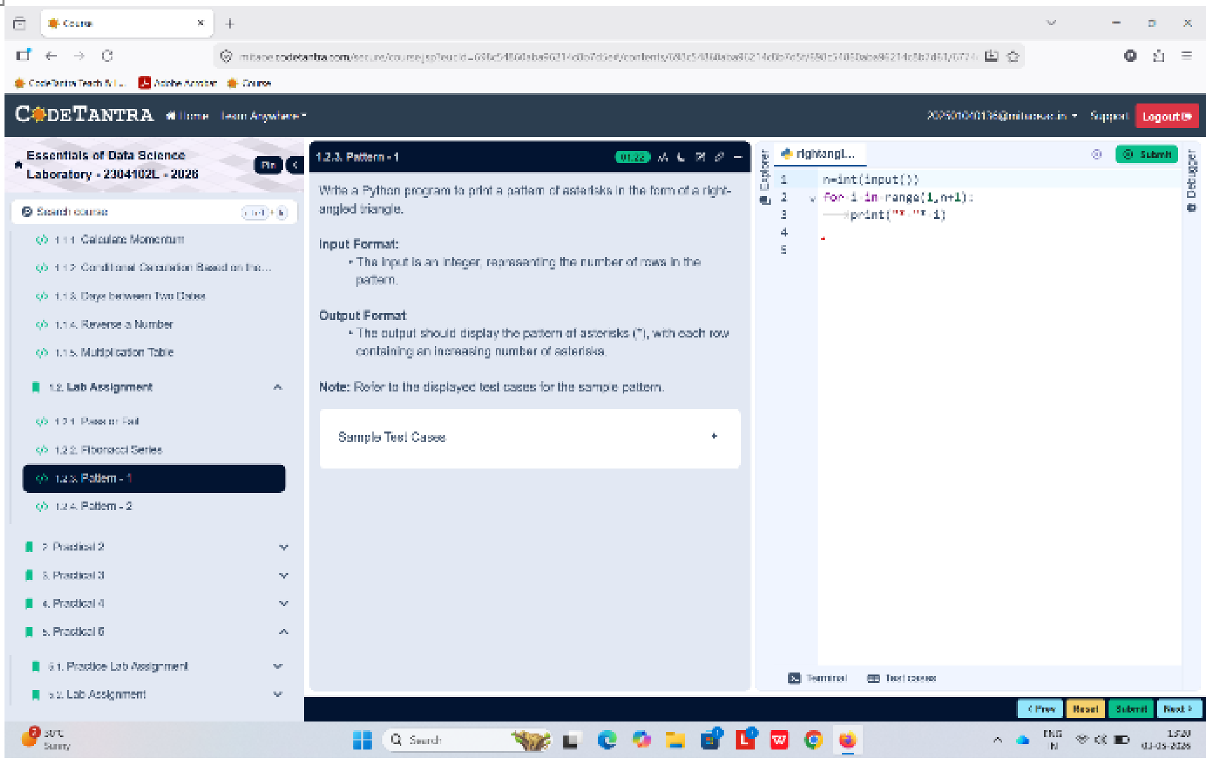
5.1. Practice Lab Assignment
1.2.2. Fibonacci Series
Write a Python program that uses recursion to print the first n terms of the Fibonacci series.
Input Format:
• A single integer n representing the number of terms to generate.
Output Format:
• A single line containing the first n terms of the Fibonacci sequence, separated by spaces.
Sample Test Cases
Setting up environment...
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share
Search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5.1. Practice Lab Assignment
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1.2.3. Pattern - 1
Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.
Input Format:
• The input is an integer, representing the number of rows in the pattern.
Output Format
• The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.
Note: Refer to the displayed test cases for the sample pattern.
Sample Test Cases
n=int(input())
for i in range(1,n+1):
print("* "*i)
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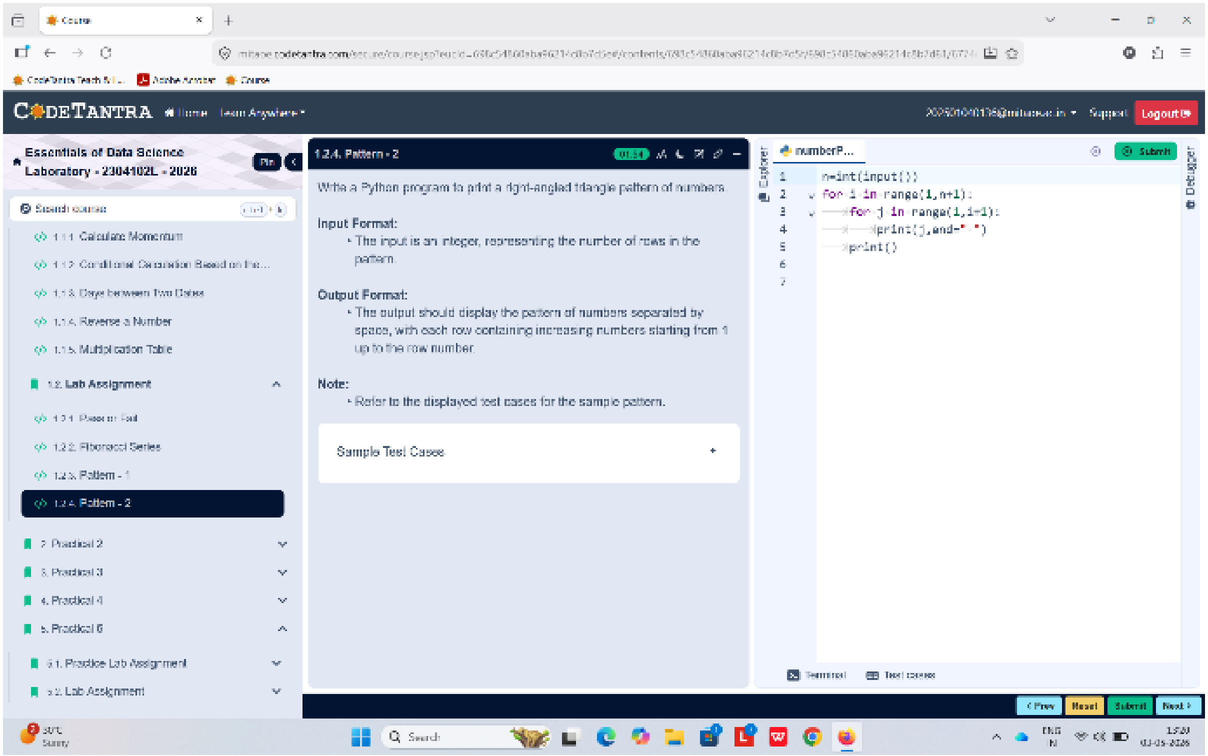
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5.1. Practice Lab Assignment
5.2. Lab Assignment
1.2.4. Pattern - 2
Write a Python program to print a right-angled triangle pattern of numbers.
Input Format:
• The input is an integer, representing the number of rows in the pattern.
Output Format:
• The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.
Note:
• Refer to the displayed test cases for the sample pattern.
Sample Test Cases
n=int(input())
for i in range(1,n+1):
for j in range(1,i+1):
print(j,end=" ")
print()
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